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Semiconductor apparatus and manufacturing method for semiconductor apparatus

Abstract

Provided is a semiconductor apparatus which can achieve high reliability while reducing parasitic capacitance and can prevent a gate leakage defect while suppressing an increase in manufacturing costs. The semiconductor apparatus includes a substrate having a buried insulating film and a semiconductor layer provided on the buried insulating film and formed with a semiconductor element, a gate electrode provided on the semiconductor layer and having a wiring extending from a central portion of the semiconductor layer to each end portion of the semiconductor layer in a case where the substrate is viewed from above, a source contact via and a drain contact via which are provided on the semiconductor layer, and a protruding portion provided near an area where the end portion of the semiconductor layer intersects the wiring of the gate electrode, the protruding portion including a material identical to that of the semiconductor layer and protruding outward from a side surface of the semiconductor layer. At least part of the end portion of the semiconductor layer has thick film regions each having a larger film thickness than a portion of the semiconductor layer immediately below the gate electrode. The source contact via and the drain contact via are provided in areas other than the thick film regions.

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Background/Summary

CROSS REFERENCE TO RELATED APPLICATIONS
(1) This application is a U.S. National Phase of International Patent Application No. PCT/JP2021/019040 filed on May 19, 2021, which claims priority benefit of Japanese Patent Application No. JP 2020-119898 filed in the Japan Patent Office on Jul. 13, 2020. Each of the above-referenced applications is hereby incorporated herein by reference in its entirety.
TECHNICAL FIELD
(2) The present disclosure (present technology) relates to a semiconductor apparatus and a manufacturing method for the semiconductor apparatus.
BACKGROUND ART
(3) Communication apparatuses for use in wireless communication and the like are provided with a high-frequency antenna switch that switches among high-frequency communication signals. Such a high-frequency antenna switch is required to be a device having an insignificant parasitic capacitance and having device characteristics that are not degraded even in a case where the antenna switch handles high frequency signals.
(4) Thus, in the related technique, for antenna switch devices, compound semiconductors such as GaAs (gallium arsenic) which have excellent high-frequency characteristics have been used. However, such compound semiconductor devices are expensive, and a device for a peripheral circuit for operating the compound semiconductor device is produced on another chip different from a chip on which the compound semiconductor device is produced. This leads to difficulty in suppressing an increase in manufacturing costs when the compound semiconductor device is incorporated into a module or the like.
(5) Thus, in recent years, antenna switch ICs (Integrated Circuits) that use an SOI (SiliconOnInsulator) substrate enabling a device for an antenna switch and a device for a peripheral circuit to be mixedly formed on an identical chip have actively been developed. The SOI substrate refers to a substrate including a buried insulating film (BOX layer) provided on a high-resistance support substrate and a semiconductor layer (SOI layer) provided on the buried insulating film and including silicon. The use of such an SOI substrate enables a reduction in parasitic capacitance due to a depletion layer formed in a PN junction region, allowing production of a device for an antenna switch that has high-frequency characteristics less likely to be degraded and that has device characteristics equal to those of the above-described compound semiconductor. Further, in a case where a device for an antenna switch is formed with use of such an SOI substrate, a device for a peripheral circuit can mixedly be formed on the identical substrate. Note that an example of the device formed on such an SOI substrate can include a semiconductor apparatus disclosed in PTL 1 listed below.
(6) However, the SOI layer may locally be thinned due to a treatment such as thermal oxidation or etching and polishing which is performed in a manufacturing process for a semiconductor apparatus. In such a thinned portion of the SOI layer, field concentration occurs during operation of a field effect transistor (hereinafter referred to as the FET), causing the reliability of the FET to be degraded. In addition, various measures have been taken to prevent such degradation of reliability of the FET, but these measures have sometimes increased the parasitic capacitance, degrading the high-frequency characteristics of the FET and significantly increasing the manufacturing costs.
(7) To deal with these problems, a technology described in PTL 2 listed below has been laid open

to public inspection, the technology having achieved low parasitic capacitance, high reliability, and low costs by increasing a film thickness of only end portions of the SOI layer.
CITATION LIST
Patent Literature
(8) [PTL 1]
(9) JP 2000-216391A
[PTL 2] JP 2018-148123A
SUMMARY
Technical Problem
(10) However, even with the above-described technology, the FET with an increased size causes CMP (Chemical Mechanical Polishing) damage on a silicon (Si) layer when STI (Shallow Trench Isolation) is formed, leading to a gate leakage defect.
(11) In view of these circumstances, an object of the present disclosure is to provide a semiconductor apparatus that can achieve high reliability while reducing the parasitic capacitance and can prevent a gate leakage defect while suppressing an increase in manufacturing costs and to provide a manufacturing method for the semiconductor apparatus.
Solution to Problem
(12) An aspect of the present disclosure provides a semiconductor apparatus including a substrate having a buried insulating film and a semiconductor layer provided on the buried insulating film and formed with a semiconductor element, a gate electrode provided on the semiconductor layer and having a wiring extending from a central portion of the semiconductor layer to each end portion of the semiconductor layer in a case where the substrate is viewed from above, a source contact via and a drain contact via which are provided on the semiconductor layer, and a protruding portion provided near an area where the end portion of the semiconductor layer intersects the wiring of the gate electrode, the protruding portion including a material identical to that of the semiconductor layer and protruding outward from a side surface of the semiconductor layer, in which at least part of the end portion of the semiconductor layer has thick film regions each having a larger film thickness than a portion of the semiconductor layer immediately below the gate electrode, and the source contact via and the drain contact via are provided in areas other than the thick film regions.
(13) Another aspect of the present disclosure provides a manufacturing method for a semiconductor apparatus, the manufacturing method including forming a gate electrode on a semiconductor layer provided on a buried insulating film and formed with a semiconductor element, the gate electrode having a wiring extending from a central portion of the semiconductor layer to end portions of the semiconductor layer in a case where a substrate having the buried insulating film and the semiconductor layer is viewed from above, forming protruding portions each located near an area where the end portion of the semiconductor layer intersects the wiring of the gate electrode, the protruding portion including a material identical to that of the semiconductor layer and protruding outward from a side surface of the semiconductor layer, and setting the end portions of the semiconductor layer larger in film thickness than a portion of the semiconductor layer immediately below the gate electrode.

Description

BRIEF DESCRIPTION OF DRAWINGS
(1) FIGS. **1**A and **1**B illustrate plan views of a semiconductor apparatus according to a first embodiment of the present disclosure.
(2) FIG. **2** illustrates cross-sectional views of the semiconductor apparatus depicted in FIGS. **1**A and **1**B, the cross-sectional views being taken along lines A-A′ and a-a′.

(3) FIG. **3** is a cross-sectional view of the semiconductor apparatus depicted in FIGS. **1**A and **1**B, the cross-sectional view being taken along line B-B′.
(4) FIG. **4** is a cross-sectional view of the semiconductor apparatus depicted in FIGS. **1**A and **1**B, the cross-sectional view being taken along line C-C′.
(5) FIG. **5** is a cross-sectional view (**1**) illustrating a process in a manufacturing method for a semiconductor apparatus according to a first embodiment of the present disclosure.
(6) FIG. **6** is a cross-sectional view (**2**) illustrating a process in the manufacturing method for the semiconductor apparatus according to the first embodiment of the present disclosure.
(7) FIG. **7** is a cross-sectional view (**3**) illustrating a process in the manufacturing method for the semiconductor apparatus according to the first embodiment of the present disclosure.
(8) FIG. **8** is a cross-sectional view (**4**) illustrating a process in the manufacturing method for the semiconductor apparatus according to the first embodiment of the present disclosure.
(9) FIG. **9** is a cross-sectional view (**5**) illustrating a process in the manufacturing method for the semiconductor apparatus according to the first embodiment of the present disclosure.
(10) FIG. **10** is a cross-sectional view (**6**) illustrating a process in the manufacturing method for the semiconductor apparatus according to the first embodiment of the present disclosure.
(11) FIGS. **11**A and **11**B depict a cross-sectional view FIG. **11**A and a plan view FIG. **11**B (**7**) illustrating a process in the manufacturing method for the semiconductor apparatus according to the first embodiment of the present disclosure.
(12) FIG. **12** is a cross-sectional view (**8**) illustrating each process in the manufacturing method for the semiconductor apparatus according to the first embodiment of the present disclosure.
(13) FIG. **13** is a cross-sectional view (**9**) illustrating each process in the manufacturing method for the semiconductor apparatus according to the first embodiment of the present disclosure.
(14) FIG. **14** is a cross-sectional view (**10**) illustrating each process in the manufacturing method for the semiconductor apparatus according to the first embodiment of the present disclosure.
(15) FIG. **15** is a cross-sectional view (**11**) illustrating each process in the manufacturing method for the semiconductor apparatus according to the first embodiment of the present disclosure.
(16) FIG. **16** is a cross-sectional view (**12**) illustrating each process in the manufacturing method for the semiconductor apparatus according to the first embodiment of the present disclosure.
(17) FIG. **17** is a cross-sectional view (**13**) illustrating each process in the manufacturing method for the semiconductor apparatus according to the first embodiment of the present disclosure.
(18) FIG. **18** is a cross-sectional view (**14**) illustrating each process in the manufacturing method for the semiconductor apparatus according to the first embodiment of the present disclosure.
(19) FIG. **19** is a cross-sectional view (**15**) illustrating each process in the manufacturing method for the semiconductor apparatus according to the first embodiment of the present disclosure.
(20) FIG. **20** is a cross-sectional view (**16**) illustrating each process in the manufacturing method for the semiconductor apparatus according to the first embodiment of the present disclosure.
(21) FIG. **21** is a plan view of a semiconductor apparatus according to Comparative Example.
(22) FIG. **22** is a plan view of a semiconductor apparatus according to a first modified example of the first embodiment of the present disclosure.
(23) FIGS. **23**A, **23**B, **23**C, and **23**D illustrate plan views (a), (b), (c), and (d) of semiconductor apparatuses according to a second modified example of the first embodiment of the present disclosure.
(24) FIG. **24** is a plan view of a semiconductor apparatus according to a second embodiment of the present disclosure.
(25) FIG. **25** is a cross-sectional view of the semiconductor apparatus depicted in FIG. **24**, the cross-sectional view being taken along line A-A′.
(26) FIG. **26** is a plan view of a semiconductor apparatus according to a third embodiment of the present disclosure.
(27) FIG. **27** is a cross-sectional view of the semiconductor apparatus depicted in FIG. **26**, the

cross-sectional view being taken along line A-A′.
(28) FIG. **28** is a cross-sectional view of the semiconductor apparatus depicted in FIG. **26**, the cross-sectional view being taken along line B-B′.
(29) FIG. **29** is a plan view of a semiconductor apparatus according to a fourth embodiment of the present disclosure.
(30) FIG. **30** is a cross-sectional view of the semiconductor apparatus depicted in FIG. **29**, the cross-sectional view being taken along line A-A′.
(31) FIG. **31** is a cross-sectional view of the semiconductor apparatus depicted in FIG. **29**, the cross-sectional view being taken along line B-B′.
DESCRIPTION OF EMBODIMENTS
(32) Embodiments of the present disclosure will be described below with reference to the drawings. In the illustration of the drawings referred to in the description below, identical or similar portions are denoted by identical or similar reference signs, and duplicate descriptions are omitted. However, it should be noted that the drawings are schematic and that the relation between thickness and planar dimensions, the ratio of the thicknesses of apparatuses and members, and the like differ from the real relation, ratio, and the like. Consequently, specific thicknesses and dimensions should be determined in consideration of the description below. In addition, it is obvious that some of the dimensional relation and the ratios vary among the drawings.
(33) In the present specification, a “first conductivity type” refers to one of a p-type or an n-type, whereas a “second conductivity type” means one of the p-type or the n-type that is different from the “first conductivity type.” In addition, “+” or “−” added to “n” and “p” means that a semiconductor region with “+” or “−” has a high or low impurity density relative to a semiconductor region without “+” or “−.” However, semiconductor regions with the same “n” do not mean that the semiconductor regions have exactly the same impurity density.
(34) In addition, the definition of directions such as up and down in the following description is only for convenience of description and is not intended to limit the technical concepts of the present disclosure. For example, it is obvious that observing a target by rotating the target through 90° converts the up-down direction into the lateral direction and observing the target by rotating the target through 180° inverts the up-down direction.
(35) Note that effects described herein are only illustrative and are not restrictive and that the present disclosure may produce other effects.
First Embodiment
(36) <Planar Configuration>
(37) FIGS. **1**A and **1**B are plan views of a semiconductor apparatus **10** according to a first embodiment of the present disclosure. Note that, in the present first embodiment described below, a transistor **12** is assumed to be an n-type MOS-FET and to have an H-shaped gate electrode structure as a planar structure. However, the transistor **12** in the present first embodiment is not limited to such an example and may be a transistor having another configuration.
(38) In the semiconductor apparatus **10** according to the present first embodiment, a buried insulating film **200** (see FIGS. **2** to **4**) is provided on a silicon support substrate **100** including high-resistance silicon with a resistivity of 500 Ωcm or more, and an impurity diffusion layer (semiconductor layer) **300** is provided on the buried insulating film **200**.
(39) The impurity diffusion layer **300** is provided with the transistor **12**. Specifically, as depicted in FIG. **1**A, The impurity diffusion layer **300** has two gate electrode wiring portions **600** and a gate electrode portion **602** constituting a gate electrode, a source electrode **800***a*, a drain electrode **800***b*, and a body contact electrode **800***c* provided thereon. The two gate electrode wiring portions **600** and the gate electrode portion **602** provided on the impurity diffusion layer **300** include polysilicon, and have a laterally brought-down H shape as viewed from above the silicon support substrate **100**. Specifically, the gate electrode portion **602** is a rectangular electrode portion located in the center in FIG. **1**A and extending along an up-down direction in FIG. **1**A. Further, the two gate electrode

wiring portions **600** are two band-like wiring portions extending in the lateral direction in FIG. **1**A and interposing the gate electrode portion **602** between the gate electrode wiring portions **600** in the up-down direction in FIG. **1**A, the gate electrode portion **602** being located in the central portion and shaped like a rectangle. In addition, the two gate electrode wiring portions **600** are connected to the above-described gate electrode portion **602** in the center.

(40) Further, as depicted in FIG. **1**A, in the present embodiment, the gate electrode wiring portion **600** extends on the impurity diffusion layer **300** along the lateral direction in FIG. **1**A from a central portion of the impurity diffusion layer **300**. The gate electrode wiring portion **600** further extends across an end portion **301** of the transistor **12** along each of the rightward direction and the leftward direction in FIG. **1**A to form a gate electrode wiring lead-out portion **604**.

(41) In addition, the source electrode **800***a* and the drain electrode **800***b*, each including a metal film, are provided interposing the gate electrode portion **602** that is located in the center of the impurity diffusion layer **300** between the source electrode **800***a* and the drain electrode **800***b* in the lateral direction. The source electrode **800***a* and the drain electrode **800***b* function as wires connected to a source region and a drain region of the transistor **12**.

(42) Then, the impurity diffusion layer **300** includes a silicon layer into which desired impurities are doped. Specifically, as depicted in FIG. **1**B, a source N+ region **305***a* and a drain N+ region **305***b* in which n-type impurities such as phosphorus or arsenic are diffused are formed below and around the source electrode **800***a* and the drain electrode **800***b* of the impurity diffusion layer **300**, and p-type impurities such as boron are diffused in the other areas of the impurity diffusion layer **300**.

(43) In addition, as depicted in the lower right of FIG. **1**A, the lower right of the impurity diffusion layer **300** is provided with the body contact electrode **800***c*. To suppress a substrate floating effect, the body contact electrode **800***c* is used as a wire for fixing and controlling a potential of the impurity diffusion layer **300**.

(44) In addition, an STI **204** (see FIGS. **2** to **4**) in which an insulating film such as a silicon oxide film is buried is provided surrounding the periphery of the impurity diffusion layer **300**. The STI **204** isolates the transistor **12** provided in the impurity diffusion layer **300** from other elements provided on the silicon support substrate **100**.

(45) In the present first embodiment, two protruding portions **350***a* and **350***b* are formed near a portion in which the gate electrode wiring portion **600** intersects the end portion **301** of the transistor **12** of the impurity diffusion layer **300**, the protruding portions **350***a* and **350***b* each including a part of the impurity diffusion layer **300** and protruding outward from a side surface of the impurity diffusion layer **300** including a silicon layer **320** of the semiconductor substrate.

(46) <Cross-Sectional Configuration>

(47) Now, a cross-sectional configuration of the semiconductor apparatus **10** in the present first embodiment will be described with reference to FIGS. **2** to **4**. FIG. **2** illustrates cross-sectional views of the semiconductor apparatus **10** depicted in FIG. **1**A, the cross-sectional views being taken along lines A-A′ and a-a′. FIG. **3** is a cross-sectional view of the semiconductor apparatus **10** depicted in FIG. **1**A, the cross-sectional view being taken along line B-B′. Further, FIG. **4** is a cross-sectional view of the semiconductor apparatus **10** depicted in FIG. **1**A, the cross-sectional view being taken along line C-C′.

(48) As depicted in FIG. **2**, the semiconductor apparatus **10** according to the present first embodiment includes the buried insulating film **200** including a silicon oxide film and provided on the silicon support substrate **100** including high-resistance silicon, as described above. Further, the semiconductor apparatus **10** includes the impurity diffusion layer **300** including a silicon layer and provided on the buried insulating film **200**. In other words, in the present first embodiment, the above-described SOI substrate is used as a substrate, and the impurity diffusion layer **300** corresponds to the above-described SOI layer. In the present first embodiment, the use of the SOI substrate enables a parasitic capacitance in the transistor **12** to be reduced. Note that a back surface

of the silicon support substrate **100** may be ground to form the silicon support substrate **100** into a thin film after manufacture of the semiconductor apparatus **10** and that the film thickness of the silicon support substrate **100** is not particularly limited. In addition, the buried insulating film **200** has a film thickness of approximately from 100 nm to 2000 nm and preferably has a film thickness of approximately 400 nm in consideration of the high frequency property of the transistor **12**.

(49) As depicted in FIG. **2**, the impurity diffusion layer **300** includes a thin film portion **300***a* located in a central portion of the impurity diffusion layer **300** and having a small film thickness, and thick film portions **300***b* located at the end portions **301** of the transistor **12** and having a larger film thickness than the thin film portion **300***a*. In addition, the impurity diffusion layer **300** includes the thick film portion **300***b* in each of the protruding portions **350***a* and **350***b* (only the protruding portion **350***b* is depicted in FIG. **2**), the thick film portion **300***b* having the same film thickness as the end portion **301** of the transistor **12**. The protruding portions **350***a* and **350***b* protrude, for example, 0.4 μm or more (denoted by ΔW in FIG. **2**) from the end portion **301** of the transistor **12** of the impurity diffusion layer **300**.

(50) The central portion of the thin film portion **300***a* located below the gate electrode portion **602** corresponds to a gate region **302** of the transistor **12** in which p-type impurities are diffused. In addition, areas of the thin film portion **300***a* interposing the gate region **302** between the areas correspond to a source region **304***a* and a drain region **304***b* in each of which n-type impurities are diffused. Note that portions of the source region **304***a* and the drain region **304***b* which are adjacent to the gate region **302** desirably form an LDD layer **2000** having a lower impurity concentration than portions of the source region **304***a* and the drain region **304***b* which are located away from the gate region **302**. This will be described in detail with reference to FIG. **19**.

(51) In addition, the thick film portion **300***b* located at the end portion **301** of the impurity diffusion layer **300** is thicker than the thin film portion **300***a* located at the central portion of the impurity diffusion layer **300**, and specifically has a film thickness twice to ten times as large as the film thickness of the thin film portion **300***a*. More specifically, in view of compatibility between the high frequency property and the reliability of the transistor **12**, the thick film portion **300***b* preferably has a film thickness of 140 nm to 200 nm, and the thin film portion **300***a* preferably has a film thickness of 20 nm to 70 nm.

(52) Note that, in the plan view of FIG. **1**A described above, the thin film portion located in the central portion of the impurity diffusion layer **300** is denoted by **300***a*, whereas the thick film portion located at the end portion **301** of the impurity diffusion layer **300** is denoted by **300***b*.

(53) Further, in the semiconductor apparatus **10** according to the present embodiment, a gate insulating film **500** is provided on the gate region **302** provided in the central portion of the impurity diffusion layer **300**. The gate insulating film **500** includes a silicon oxide film, and any film thickness can be selected for the gate insulating film **500**.

(54) In addition, two silicide films **702** spaced apart from the gate region **302** are provided on respective upper surfaces of the impurity diffusion layer **300** located on both sides of the gate region **302**. Further, a source contact via **700***a* and the source electrode **800***a* are provided on one of the silicide films **702**, and a drain contact via **700***b* and the drain electrode **800***b* are provided on the other of the silicide films **702**. In other words, the source contact via **700***a* and the drain contact via **700***b* are respectively provided on the source region **304***a* and the drain region **304***b* formed in the thin film portion **300***a* of the impurity diffusion layer **300**. The gate electrode portion **602** is provided being interposed between the source region **304***a* and the drain region **304***b*. The source contact via **700***a* and the drain contact via **700***b* provided on the thin film portion **300***a* enable the parasitic capacitance between the source and the drain to be reduced. Note that the silicide film **702** is a compound film including silicon and other elements and that the source contact via **700***a*, the drain contact via **700***b*, the source electrode **800***a*, and the drain electrode **800***b* each include a metal film or the like. Note that, in the present first embodiment, for the source region **304***a*, the drain region **304***b*, the silicide film **702**, the source contact via **700***a*, the drain contact via **700***b*, the

source electrode **800***a*, and the drain electrode **800***b*, the film thickness, the size, the shape, and the like are not particularly limited. In addition, in the present first embodiment, the transistor **12** is preferably laid out in consideration of a manufacturing variation in order to maintain a high manufacturing yield of the semiconductor apparatus **10**.

(55) Note that, in the above description, to reduce the parasitic capacitance between the source and the drain, the source contact via **700***a* and the drain contact via **700***b* are provided on the thin film portion **300***a* of the impurity diffusion layer **300**.

(56) In addition, the STI (isolation insulating film) **204** is provided around the impurity diffusion layer **300** in order to isolate the transistor **12** from the other elements. In particular, the STI **204** includes a trench provided surrounding the periphery of the impurity diffusion layer **300** and a silicon oxide film buried in the trench. Note that, in the present embodiment, the width, the depth, the shape, and the like of the trench of the STI **204** are not particularly limited.

(57) In addition, the silicon oxide film **202** is provided covering the gate electrode portion **602**, the impurity diffusion layer **300**, and the STI **204**. Further, a further silicon nitride film **400** is provided covering the above-described silicon oxide film **202**. In addition, on the silicon nitride film **400**, a silicon oxide film **802** is provided between the source contact via **700***a* and the drain contact via **700***b* and between the source electrode **800***a* and the drain electrode **800***b*. Note that, in the present first embodiment, for the silicon oxide film **202**, the silicon nitride film **400**, and the silicon oxide film **802**, the material, the film thickness, and the like are not particularly limited.

(58) Now, with reference to FIG. **3**, the semiconductor apparatus **10** according to the present first embodiment will be described. As described above, also in this cross-sectional view, the semiconductor apparatus **10** includes the silicon support substrate **100**, the buried insulating film **200** provided on the silicon support substrate **100**, and the impurity diffusion layer **300** provided on the buried insulating film **200**.

(59) Also in the cross section of FIG. **3**, as is the case with the cross section of FIG. **2**, the impurity diffusion layer **300** includes the thin film portion **300***a* located at the central portion and having a small film thickness and the thick film portions **300***b* located at the respective end portions **301** of the transistor **12** and having a large film thickness. In particular, also in this cross section, the thick film portion **300***b* has a film thickness twice to ten times as large as the film thickness of the thin film portion **300***a*. More specifically, the thick film portion **300***b* preferably has a film thickness of 140 nm to 200 nm, and the thin film portion **300***a* preferably has a film thickness of 20 nm to 70 nm.

(60) In addition, in the cross section of FIG. **3**, the gate electrode wiring portion **600** is provided on the thin film portion **300***a* and the thick film portion **300***b* of the impurity diffusion layer **300** via the gate insulating film **500**. As depicted in FIG. **3**, the gate electrode wiring portion **600** extends on the impurity diffusion layer **300** in the lateral direction in FIG. **3** and further extends across the end portion **301** of the transistor **12** in each of the rightward direction and the leftward direction in FIG. **3** to form the gate electrode wiring lead-out portion **604**. In other words, the gate electrode wiring portion **600** is provided to pass over the thick film portions **300***b* of the impurity diffusion layer **300** as well as over the thin film portion **300***a* of the impurity diffusion layer **300**.

(61) Now, with reference to FIG. **4**, the semiconductor apparatus **10** according to the present first embodiment will be described. The cross section is obtained by cutting the body contact electrode **800***c* in such a manner as to traverse the body contact electrode **800***c*. As described above, also in this cross-sectional view, the semiconductor apparatus **10** includes the silicon support substrate **100**, the buried insulating film **200** provided on the silicon support substrate **100**, and the impurity diffusion layer **300** provided on the buried insulating film.

(62) In the cross section of FIG. **4** as well, as is the case with the cross section of FIG. **2**, the impurity diffusion layer **300** includes the thin film portion **300***a* located in the central portion and having a small film thickness and the thick film portions **300***b* located at the respective end portions **301** of the transistor **12** and having a large film thickness. Specifically, also in this cross

section, the thick film portion **300***b* has a film thickness twice to ten times as large as the film thickness of the thin film portion **300***a*. More specifically, the thick film portion **300***b* preferably has a film thickness of 140 nm to 200 nm, and the thin film portion **300***a* preferably has a film thickness of 20 nm to 70 nm.
(63) In addition, also in the cross section of FIG. **4**, the gate electrode wiring portions **600** are provided on the thin film portion **300***a* of the impurity diffusion layer **300** via the gate insulating film **500**.
(64) Further, as depicted in the left of FIG. **4**, a body contact via **700***c* and the body contact electrode **800***c* are provided on the thick film portion **300***b* via the silicide film **702**. Note that, as described above, the silicide film **702** is a compound film of silicon and other elements, and the body contact via **700***c* and the body contact electrode **800***c* each include a metal film or the like. Note that, in the present first embodiment, for the silicide film **702**, the body contact via **700***c*, and the body contact electrode **800***c*, the film thickness, the size, the shape, and the like are not particularly limited.
(65) As described above, the source contact via **700***a* and the drain contact via **700***b* are provided on the thin film portion **300***a* of the impurity diffusion layer **300**. This reduces the parasitic capacitance between the source and the drain. On the other hand, the body contact via **700***c* is provided on the thick film portion **300***b* of the impurity diffusion layer **300**. The parasitic capacitance between the body (impurity diffusion layer **300**) and the gate only insignificantly affects the high-frequency property of the transistor **12**, and thus the body contact via **700***c* may be provided on the thick film portion **300***b* of the impurity diffusion layer **300**.
(66) <Manufacturing Method for Semiconductor Apparatus>
(67) Now, a manufacturing method for the semiconductor apparatus **10** will be described.
(68) First, as depicted in FIG. **5**, the manufacturing method according to the present embodiment uses an SOI substrate including the silicon support substrate **100**, the buried insulating film **200** including a silicon oxide film and having a film thickness of 100 nm to 2000 nm, preferably 400 nm, and further the silicon layer **320** provided on the buried insulating film **200** and having a film thickness of 30 nm to 400 nm, preferably 175 nm.
(69) Then, as depicted in FIG. **6**, an oxidation treatment is performed on an upper surface of the silicon layer **320** to form a silicon oxide film **900** having a film thickness of 10 nm to 100 nm, preferably 10 nm. Note that a method for the above-described oxidation treatment is not particularly limited and that known various methods for an oxidation treatment can be used. Further, a silicon nitride film **902** having a film thickness of 10 nm to 300 nm, preferably 100 nm, is formed on the silicon oxide film **900** by CVD (Chemical Vapor Deposition).
(70) Then, as depicted in FIG. **7**, resist is applied all over the surface of the silicon nitride film **902**, and is exposed through photolithography to form a resist pattern **904**. The resist pattern **904** has a pattern including an opening in an area of the silicon layer **320** that is to have a reduced film thickness. The pattern is preferably a layout pattern designed in such a manner that the area where the film thickness located between the thin film portion (corresponding to the thin film portion **300***a* described above) and the thick film portion **300***b* (corresponding to the thick film portion **300***b* described above) of the silicon layer **320** gradually varies has a length of approximately 400 nm, that is, the distance between the thin film portion and the thick film portion is approximately 400 nm.
(71) Subsequently, the resist pattern **904** is used as a mask to perform a dry etching treatment on the silicon nitride film **902** and the silicon oxide film **900**. In this manner, as depicted in FIG. **8**, after an opening **906** corresponding to a partly exposed upper surface of the silicon layer **320** is obtained, the resist pattern **904** is removed. Note that, when the dry etching treatment is performed on the silicon oxide film **900**, the upper surface of the silicon layer **320** exposed from the opening **906** may be partly etched.
(72) Then, as depicted in FIG. **9**, a selective oxidation treatment (LOCal Oxidation of Silicon;

LOCOS oxidation treatment) is performed on a part of the silicon layer **320** exposed from the opening **906**. At this time, the amount of oxidation by which the silicon layer **320** is oxidized by the oxidation treatment is controlled in such a manner that the silicon layer **320** located at the opening **906** has a desired film thickness. More specifically, in a case where the film thickness of the central portion **320***a* of the silicon layer **320** (that is, the film thickness of the thin film portion **300***a* of the impurity diffusion layer **300**) is finally 60 nm, then in the process in FIG. **9**, the amount of oxidation is preferably controlled to set the film thickness of the silicon layer **320** located at the opening **906** to approximately 80 nm. This causes the silicon layer **320** to be partly thinned.
(73) Subsequently, when the silicon nitride film **902** is removed with phosphoric acid and the silicon oxide film **900** is further removed with hydrofluoric acid or the like, the silicon layer **320** can be obtained that includes a silicon thin film portion **908** corresponding to the silicon layer **320** partly thinned as depicted in FIG. **10**.
(74) Further, as depicted in FIG. **11**A, the oxidation treatment is performed on the upper surface of the silicon layer **320** to form a silicon oxide film **1000** having a film thickness of 10 nm to 100 nm, preferably 10 nm. Further, a silicon nitride film **1100** having a film thickness of 10 nm to 400 nm, preferably 210 nm is formed on the silicon oxide film **1000** by CVD. Then, resist is applied all over the surface of the silicon nitride film **1100**, and is exposed by photolithography to form a resist pattern **1300**. The resist pattern **1300** has a pattern including an opening at a position where the STI **204** for separating the transistor **12** from the other elements is formed.
(75) As depicted in FIG. **11**B, the resist pattern **1300** includes the resist pattern **1300** for element isolation including resist patterns **350***a*′ and **350***b*′ with protruding portions in such a manner that silicon that is located near the areas where the end portion **301** of the transistor **12** and the gate electrode wiring portions **600** intersect protrudes. This configuration suppresses an increase in the number of processes for photolithography.
(76) Subsequently, the dry etching treatment is performed on the silicon nitride film **1100** and the silicon oxide film **1000** with the resist pattern **1300** used as a mask. Further, the upper surface of the silicon layer **320** is exposed in areas that are not covered with the resist pattern **1300**, and then the resist pattern **1300** is removed. Note that, in the present first embodiment, a method for removing the resist pattern **1300** is not particularly limited and that known various removal methods such as asking can be used. Then, a structure depicted in FIG. **12** can be obtained by using the dry etching treatment under conditions different from those described above to etch the silicon layer **320** with the silicon nitride film **1100** used as a mask.
(77) Subsequently, as depicted in FIG. **13**, HDP (High Density Plasma) or the like is used to form a silicon oxide film **1400** entirely above the silicon support substrate **100** in such a manner that trenches respectively provided on both sides of the silicon layer **320** are filled with the silicon oxide film **1400**. At this time, the silicon oxide film **1400** may be formed to cover an upper surface of the silicon nitride film **1100**, and the silicon oxide film **1400** has a film thickness of from 50 nm to 1000 nm, preferably 400 nm.
(78) Then, resist is applied all over the surface of the silicon oxide film **1400** and is exposed by photolithography to form a resist film **1500**. The resist film **1500** has a pattern including an opening **1600** in an area corresponding to portions of the silicon oxide film **1400** and the silicon nitride film **1100** which are located on the central portion **320***a* of the silicon layer **320** and which are to be removed. At this time, the opening **1600** spreads above the central portion **320***a* of the silicon layer **320** and preferably further spreads to the thick film portions of the silicon layer **320** located at respective end portions **320***b* of the silicon layer **320** and having a large film thickness.
(79) Then, the dry etching treatment is performed on the silicon oxide film **1400** with the resist film **1500** used as a mask to remove the silicon oxide film **1400**. This allows a structure depicted in FIG. **14** to be obtained. Note that, depending on conditions for CMP (Chemical Mechanical Polishing) to be performed later, the silicon oxide film **1400** may remain in the central portion **320***a* and in an area between the central portion **320***a* and the end portion **320***b* where the film thickness of the

silicon layer **320** gradually varies. Thus, to prevent the silicon oxide film **1400** from remaining, the above-described dry etching treatment is preferably performed under overetching conditions of such an extent that the silicon nitride film **1100** is also etched.
(80) Then, as depicted in FIG. **15**, the resist film **1500** is removed.
(81) Further, a structure depicted in FIG. **16** can be obtained by performing a planarization treatment on the upper surface of the silicon support substrate **100** by CMP. Note that the planarized silicon oxide film **1400** forms the STI **204** for element isolation.
(82) Subsequently, the silicon nitride film **1100** is removed with phosphoric acid, and the silicon oxide film **1000** is further removed with hydrofluoric acid or the like. Then, a transistor formation region **1700** as depicted in FIG. **17** can be obtained. Here, for example, ion implantation may be used to dope impurities into the transistor formation region **1700** as necessary. At this time, an upper surface of the transistor formation region **1700** is covered with patterned resist, and impurities may be doped into a desired portion of the transistor formation region **1700**.
(83) Further, the gate insulating film **500** including a silicon oxide film is formed on the transistor formation region **1700** and the STI **204**. In addition, as depicted in FIG. **18**, a polysilicon film is formed all over the surface of the gate insulating film **500**, and is patterned into any shape by etching or the like to form the gate electrode portion **602**.
(84) Subsequently, ion implantation may be used to dope impurities into the transistor formation region **1700** with the gate electrode portion **602** used as a mask to form the impurity diffusion layer **300**. Further, impurities are doped around the gate region **302** of the impurity diffusion layer **300** to achieve an impurity concentration lower than that achieved with the above-described ion implantation, thus forming an LDD (Lightly Doped Drain) layer **2000** in the impurity diffusion layer **300**. This allows a structure depicted in FIG. **19** to be obtained. Note that the above-described ion implantation may be performed after formation of the LDD layer **2000**.
(85) Further, etching is performed with the gate electrode portion **602** used as a mask to pattern the gate insulating film **500**. Subsequently, the silicide films **702** may be formed on the exposed upper surface of the impurity diffusion layer **300** at positions spaced apart from the gate electrode portion **602** and on both sides of the gate electrode portion **602**. Note that, in the present embodiment, a method for forming the above-described silicide films **702** is not particularly limited and that known various formation methods can be used.
(86) Subsequently, the silicon oxide film **202**, the silicon nitride film **400**, and the silicon oxide film **802** are sequentially formed on the impurity diffusion layer **300**, the STI **204**, and the gate electrode portion **602**. Then, the source contact via **700***a* and the drain contact via **700***b* each of which extends from the silicon oxide film **802**, penetrates the silicon nitride film **400** and the silicon oxide film **202**, and reaches the silicide film **702** are formed. At this time, in the present embodiment, the source contact via **700***a* and the drain contact via **700***b* can be provided on a wide thin film portion **300***a* away from each other at a predetermined distance. This enables accurate patterning of the source contact via **700***a* and the drain contact via **700***b* to be avoided, allowing a decrease in manufacturing yield to be avoided. In addition, since the source contact via **700***a* and the drain contact via **700***b* can be spaced apart from each other at the predetermined distance, particularly in a transistor with a plurality of gates, an increase in the layout size of the transistor can be suppressed, thus allowing an increase in manufacturing costs to be suppressed.
(87) Further, the source electrode **800***a* is formed on the source contact via **700***a*, and the drain electrode **800***b* is formed on the drain contact via **700***b*. At this time, methods for forming the silicon oxide film **202**, the silicon nitride film **400**, the silicon oxide film **802**, the source contact via **700***a*, the drain contact via **700***b*, the source electrode **800***a*, and the drain electrode **800***b* are not particularly limited, and formation methods commonly used in a manufacturing method for the semiconductor apparatus **10** can be used. Further, a further metal film may be formed on the source electrode **800***a* and the drain electrode **800***b*. As described above, the semiconductor apparatus **10** according to the embodiment of the present disclosure depicted in FIGS. **1**A, **1**B, **2**, **3**, and **4** can be

obtained.
Comparative Example of Embodiment
(88) FIG. **21** depicts a plan view of the semiconductor apparatus **10** as Comparative Example. Note that identical portion in FIGS. **21**, **1**A, and **1**B described above are denoted by identical reference signs, and detailed descriptions of these portions are omitted.
(89) In Comparative Example, the film thickness of the impurity diffusion layer **300** is increased only at the end portions **301** of the impurity diffusion layer **300**, thus achieving low parasitic capacitance, high reliability, and low costs.
(90) However, an increased size of the transistor **12** poses a problem in that physical damage (microcracks) is likely to be caused to the impurity diffusion layer **300** of the gate electrode wiring lead-out portion **604** in the CMP process during STI formation. The physical damage degrades the film quality of the gate insulating film **500** in the subsequent thermal oxidation process and reduces withstand voltage for electric fields, leading to a gate leakage defect.
Solution of First Embodiment
(91) Thus, in the present first embodiment, as depicted in FIG. **1**A, the two protruding portions **350***a* and **350***b* are formed near the portion in which the gate electrode wiring portion **600** intersects the end portion **301** of the impurity diffusion layer **300**, the protruding portions **350***a* and **350***b* protruding outward from the side surface of the impurity diffusion layer **300** including the silicon layer **320** of the semiconductor substrate.
(92) The two protruding portions **350***a* and **350***b* each have a length of 0.4 μm or more in a direction parallel to the extension direction of the gate electrode wiring portions **600** and a length of 0.7 μm or more in a direction perpendicular to the extension direction. Each protruding portion preferably has a length of 1 μm or more in the direction parallel to the extension direction and a length of 1 μm or more in the direction perpendicular to the extension direction.
(93) In addition, each of the two protruding portions **350***a* and **350***b* has a distance of 14 nm or more and 200 nm or less to the gate electrode wiring lead-out portion **604**.
Effects of First Embodiment
(94) As described above, according to the above-described first embodiment, the two protruding portions **350***a* and **350***b* that interpose the gate electrode wiring lead-out portion **604** therebetween are formed. Formation of the protruding portions **350***a* and **350***b* provides the end portion **301** of the impurity diffusion layer **300** with a corresponding additional area and a corresponding additional film thickness. This protects, from CMP damage, the area where the end portion **301** of the impurity diffusion layer **300** intersects the gate electrode wiring portion **600**, preventing gate leakage. Thus, high reliability and high yield are achieved to enable the manufacturing costs to be reduced.
(95) In addition, according to the first embodiment, the distance between each of the protruding portions **350***a* and **350***b* and the gate electrode wiring lead-out portion **604** is 14 nm or more and 200 nm or less. Thus, the material of the gate electrode wiring lead-out portion **604** does not remain in the protruding portions **350***a* and **350***b*. This prevents an increase in unwanted gate parasitic capacitance and suppresses dusting to achieve high reliability and high yield, allowing an increase in manufacturing costs to be suppressed.
First Modified Example of First Embodiment
(96) FIG. **22** is a plan view of a semiconductor apparatus **10**A as a first modified example of the first embodiment of the present disclosure. Identical portion in FIG. **22** and described above are denoted by identical reference signs, and detailed descriptions of these portions are omitted.
(97) A transistor **12**A forms the gate electrode portion **602** and the gate electrode wiring lead-out portion **604** corresponding to a lead-out portion of the gate electrode wiring portion **600**. In addition, the transistor **12**A forms one protruding portion **350***a* near the portion in which the gate electrode wiring portion **600** intersects the end portion **301** of the transistor **12**A, the protruding portion **350***a* protruding outward from the side surface of the impurity diffusion layer **300** including

the silicon layer **320** of the semiconductor substrate. In the first modified example, in a case where two or more gate electrode wiring lead-out portions **604** are disposed parallel to each other, the protruding portion **350***a* is formed opposite to end portions of the gate electrode wiring lead-out portions **604** facing each other, in other words, formed outside the gate electrode wiring lead-out portions **604**.

(98) The first modified example as described above also produces effects similar to those of the above-described first embodiment.

Second Modified Example of First Embodiment

(99) FIGS. **23**A, **23**B, **23**C and **23**D is a are plan views of a semiconductor apparatus **10**B as a second modified example of the first embodiment of the present disclosure. Identical portion in FIGS. **23**A, **23**B, **23**C, **23**D, and described above are denoted by identical reference signs, and detailed descriptions of these portions are omitted.

(100) In FIG. **23**A, a transistor **12**B forms two protruding portions **351***a* and **351***b* near the portion in which the gate electrode wiring portion **600** intersects the end portion **301** of the transistor **12**B, the protruding portions **351***a* and **351***b* protruding outward from the side surface of the impurity diffusion layer **300** including the silicon layer **320** of the semiconductor substrate. The two protruding portions **351***a* and **351***b* have a round shape as viewed from above the transistor **12**B.

(101) In addition, in FIG. **23**B, the two protruding portions **352***a* and **352***b* are triangles as viewed from above the transistor **12**B. In addition, in FIG. **23**C, the two protruding portions **353***a* and **353***b* are triangles having different orientations from those in FIG. **23**B. Further, in FIG. **23**D, the two protruding portions **354***a* and **354***b* are polygons as viewed from above the transistor **12**B.

(102) The second modified example as described above also produces effects similar to those of the above-described first embodiment.

Second Embodiment

(103) FIG. **24** is a plan view of a semiconductor apparatus **10**C as a second embodiment of the present disclosure. Identical portion in FIGS. **23**A, **23**B, **23**C, **23**D, and in FIG. **1**A described above are denoted by identical reference signs, and detailed descriptions of these portions are omitted. FIG. **25** is a cross-sectional view of the semiconductor apparatus **10**C depicted in FIG. **24**, the cross-sectional view being taken along line A-A′. Identical portions in FIG. **25** and in FIG. **2** described above are denoted by identical reference signs, and detailed descriptions of these portions are omitted.

(104) As depicted in FIG. **24**, the semiconductor apparatus **10**C and a transistor **12**C include a plurality of gate electrode portions **602** shaped like a ladder as viewed from above the silicon support substrate **100** and two gate electrode wiring portions **600**A. In particular, the plurality of gate electrode portions **602** is arranged along the lateral direction in FIG. **24**. The two gate electrode wiring portions **600**A interpose the plurality of gate electrode portions **602** therebetween along the up-down direction in FIG. **24** to connect the plurality of gate electrode portions **602** together. Further, the source electrodes **800***a* and the drain electrodes **800***b* are provided in such a manner that each source electrode **800***a* and each drain electrode **800***b* interpose each gate electrode portion **602** therebetween in the lateral direction in FIG. **24**.

(105) In addition, the semiconductor apparatus **10**C and the transistor **12**C form two protruding portions **350***a* and **350***b* near the portion in which the gate electrode wiring portion **600**A intersects the end portion **301** of the transistor **12**C, the protruding portions **350***a* and **350***b* protruding outward from the side surface of the impurity diffusion layer **300** including the silicon layer **320** of the semiconductor substrate. In addition, the gate electrode wiring portion **600**A forms the gate electrode wiring lead-out portion **604** extending across the end portion **301** of the transistor **12**C.

(106) In addition, also in the present second embodiment, as depicted in FIG. **25**, the impurity diffusion layer **300** includes the thin film portion **300***a* located in the central portion and the thick film portions **300***b* located at the end portions **301** as is the case with the above-described first embodiment. In the present second embodiment as well, the gate region **302**, the source region

304*a*, and the drain region **304***b* are provided in the thin film portion **300***a* of the impurity diffusion layer **300**. As described above, even with the plurality of gate regions **302**, the impurity diffusion layer **300** with the thin film portion **300***a* and the thick film portion **300***b* can be applied.

(107) The second embodiment as described above also produces effects similar to those of the above-described first embodiment.

Third Embodiment

(108) FIG. **26** depicts a plan view of a semiconductor apparatus **10**D as a third embodiment of the present disclosure. Identical portion in FIG. **26** and in FIG. **1**A described above are denoted by identical reference signs, and detailed descriptions of these portions are omitted. In addition, FIG. **27** is a cross-sectional view of the semiconductor apparatus **10**D depicted in FIG. **26**, the cross-sectional view being taken along A-A′. Identical portions in FIG. **27** and in FIG. **2** described above are denoted by identical reference signs, and detailed descriptions of these portions are omitted. Further, FIG. **28** is a cross-sectional view of the semiconductor apparatus **10**D depicted in FIG. **26**, the cross-sectional view being taken along B-B′. Identical portions in FIG. **28** and in FIG. **4** described above are denoted by identical reference signs, and detailed descriptions of these portions are omitted.

(109) As depicted in FIG. **26**, the semiconductor apparatus **10**D according to the third embodiment includes a gate electrode wiring portion **600**B and the gate electrode portion **602**A that form a T shape as viewed from above the silicon support substrate **100**. In particular, the gate electrode portion **602**A is shaped like a rectangle extending along the up-down direction in FIG. **26**. The gate electrode wiring portion **600**B is a rectangular wiring portion extending along the lateral direction in FIG. **26**. In addition, the source electrode **800***a* and the drain electrode **800***b* are provided interposing the gate electrode portion **602**A therebetween in the lateral direction in FIG. **26**.

(110) In addition, the semiconductor apparatus **10**D and the transistor **12**D form two protruding portions **350***a* and **350***b* near the portion in which the gate electrode wiring portion **600**B intersects the end portion **301** of the transistor **12**D, the protruding portions **350***a* and **350***b* protruding outward from the side surface of the impurity diffusion layer **300** including the silicon layer **320** of the semiconductor substrate. Further, the semiconductor apparatus **10**D and the transistor **12**D form two protruding portions **360***a* and **360***b* near the portion in which the gate electrode portion **602**A intersects the end portion **301** of the transistor **12**D, the protruding portions **360***a* and **360***b* protruding outward from the side surface of the impurity diffusion layer **300** including the silicon layer **320** of the semiconductor substrate. In addition, the gate electrode portion **602**A forms a gate electrode lead-out portion **605** extending across the end portion **301** of the transistor **12**D.

(111) In addition, also in the present third embodiment, as depicted in FIG. **27**, the impurity diffusion layer **300** includes the thin film portion **300***a* located in the central portion and the thick film portions **300***b* located at the end portions **301** as is the case with the above-described first embodiment. In the present third embodiment as well, the gate region **302**, the source region **304***a*, and the drain region **304***b* are provided in the thin film portion **300***a* of the impurity diffusion layer **300**. As described above, even with the gate electrode wiring portion **600**B and the gate electrode portion **602**A forming a T shape, the impurity diffusion layer **300** with the thin film portion **300***a* and the thick film portion **300***b* can be applied.

(112) The third embodiment as described above also produces effects similar to those of the above-described first embodiment.

Fourth Embodiment

(113) FIG. **29** depicts a plan view of a semiconductor apparatus **10**E as a fourth embodiment of the present disclosure. Identical portions in FIG. **29** and in FIG. **26** described above are denoted by identical reference signs, and detailed descriptions of these portions are omitted. In addition, FIG. **30** is a cross-sectional view of the semiconductor apparatus **10**E depicted in FIG. **29**, the cross-sectional view being taken along A-A′. Identical portions in FIG. **30** and in FIG. **2** described above are denoted by identical reference signs, and detailed descriptions of these portions are omitted.

Further, FIG. **31** is a cross-sectional view of the semiconductor apparatus **10**E depicted in FIG. **29**, the cross-sectional view being taken along B-B′. Identical portions in FIG. **31** and in FIG. **4** described above are denoted by identical reference signs, and detailed descriptions of these portions are omitted.

(114) As depicted in FIG. **29**, the semiconductor apparatus **10**E according to the fourth embodiment includes the gate electrode portion **602**B having an I shape as viewed from above the silicon support substrate **100**. In particular, the gate electrode portion **602**B has a rectangular shape extending along the up-down direction in FIG. **29**. Further, the source electrode **800***a* and the drain electrode **800***b* are provided interposing the gate electrode portion **602**B therebetween in the lateral direction in FIG. **29**.

(115) In addition, the semiconductor apparatus **10**E and the transistor **12**E form two protruding portions **360***a* and **360***b* near the portion in which the gate electrode portion **602**B intersects the end portion **301** of the transistor **12**E, the protruding portions **360***a* and **360***b* protruding outward from the side surface of the impurity diffusion layer **300** including the silicon layer **320** of the semiconductor substrate. In addition, the gate electrode portion **602**B forms the gate electrode lead-out portion **605** extending across the end portion **301** of the transistor **12**E.

(116) In addition, in the present fourth embodiment, the impurity diffusion layer **300** includes the thin film portion **300***a* located in the central portion and the thick film portions **300***b* located at the end portions **301** as is the case with the above-described first embodiment. In the present fourth embodiment, the gate region **302**, the source region **304***a*, and the drain region **304***b* are provided in the thin film portion **300***a* of the impurity diffusion layer **300**. As described above, even with the gate electrode portion **602**B having an I shape, the impurity diffusion layer **300** with the thin film portion **300***a* and the thick film portion **300***b* can be applied.

(117) The fourth embodiment as described above also produces effects similar to those of the above-described first embodiment.

Other Embodiments

(118) As described above, the present technology has been described in conjunction with the first to fourth embodiments and the first and second modified examples of the first embodiment. However, it should not be understood that the discussions and the drawings constituting a part of the disclosure limit the present technology. By understanding the spirits of contents of the technology disclosed by the above-described first to third embodiments, those skilled in the art clearly recognize that the present technology may include various alternative embodiments, examples, and operation technologies. In addition, configurations disclosed by the first to fourth embodiments and the first and second modified examples of the first embodiment can appropriately be combined together without leading to contradiction. For example, configurations disclosed by a plurality of different embodiments may be combined together or configurations disclosed by a plurality of different modified examples of the identical embodiment may be combined together.

(119) Note that the present disclosure can also be configured as described below. (1) A semiconductor apparatus including: a substrate having a buried insulating film and a semiconductor layer provided on the buried insulating film and formed with a semiconductor element; a gate electrode provided on the semiconductor layer and having a wiring extending from a central portion of the semiconductor layer to each end portion of the semiconductor layer in a case where the substrate is viewed from above; a source contact via and a drain contact via which are provided on the semiconductor layer; and a protruding portion provided near an area where the end portion of the semiconductor layer intersects the wiring of the gate electrode, the protruding portion including a material identical to that of the semiconductor layer and protruding outward from a side surface of the semiconductor layer, in which at least part of the end portion of the semiconductor layer has thick film regions each having a larger film thickness than a portion of the semiconductor layer immediately below the gate electrode, and the source contact via and the drain contact via are provided in areas other than the thick film regions. (2) The semiconductor apparatus according to

(1) described above, in which the wiring has lead-out portions extending outward from the respective end portions of the semiconductor layer, and two or more protruding portions are provided interposing the lead-out portion of the gate electrode between the protruding portions. (3) The semiconductor apparatus according to (1) described above, in which the wiring has lead-out portions extending outward from the respective end portions of the semiconductor layer, the lead-out portions of the gate electrode are disposed in plural in such a manner that the lead-out portions are parallel to one another, and the protruding portion is provided at an end portion opposite to an end portion at which the lead-out portions of the gate electrode face each other. (4) The semiconductor apparatus according to any one of (1) to (3) described above, further including: an element isolation portion isolating the semiconductor element. (5) The semiconductor apparatus according to any one of (1) to (3) described above, in which the end portion of the semiconductor layer has a film thickness twice to ten times as large as a film thickness of the central portion of the semiconductor layer. (6) The semiconductor apparatus according to (5) described above, in which the end portion of the semiconductor layer has a film thickness of 140 nm to 200 nm, and the central portion of the semiconductor layer has a film thickness of 20 nm to 70 nm. (7) The semiconductor apparatus according to any one of (1) to (3) described above, in which the protruding portion protrudes 0.4 μm or more in a direction parallel to an extension direction of the wiring of the gate electrode and protrudes 0.7 μm or more in a direction perpendicular to the extension direction, or the protruding portion protrudes 1 μm or more in the direction parallel to the extension direction and protrudes 1 μm or more in the direction perpendicular to the extension direction. (8) The semiconductor apparatus according to (2) or (3) described above, in which a distance between the protruding portion and the lead-out portion of the gate electrode is 14 nm or more and 200 nm or less. (9) The semiconductor apparatus according to any one of (1) to (3) described above, in which the protruding portion has at least one shape of a quadrangle, a round shape, a triangle, and a polygon, as viewed from above the substrate. (10) The semiconductor apparatus according to any one of (1) to (3) described above, in which the gate electrode has any one of an H shape, a T shape, an I shape, and a ladder shape, as viewed from above the substrate. (11) A manufacturing method for a semiconductor apparatus, the manufacturing method including: forming a gate electrode on a semiconductor layer provided on a buried insulating film and formed with a semiconductor element, the gate electrode having a wiring extending from a central portion of the semiconductor layer to end portions of the semiconductor layer in a case where a substrate having the buried insulating film and the semiconductor layer is viewed from above; forming protruding portions each located near an area where the end portion of the semiconductor layer intersects the wiring of the gate electrode, the protruding portion including a material identical to that of the semiconductor layer and protruding outward from a side surface of the semiconductor layer; and setting the end portions of the semiconductor layer larger in film thickness than a portion of the semiconductor layer immediately below the gate electrode.

REFERENCE SIGNS LIST

(120) **10**, **10**A, **10**B, **10**C, **10**D, **10**E: Semiconductor apparatus **12**, **12**A, **12**B, **12**C, **12**D, **12**E: Transistor **100**: Silicon support substrate **200**: Buried insulating film **202**: Silicon oxide film **204**: STI (isolation insulating film) **400**: Silicon nitride film **802**: Silicon oxide film **300**: Impurity diffusion layer (semiconductor layer) **300***a*: Thin film portion **300***b*: Thick film portion **301**: End portion of transistor **302**: Gate region **304***a*: Source region **304***b*: Drain region **320**: Silicon layer **320***a*: Central portion **320***b*: End portion **350***a*, **350***b*, **351***a*, **351***b*, **352***a*, **352***b*, **353***a*, **353***b*, **354***a*, **354***b*, **360***a*, **360***b*: Protruding portion **500**: Gate insulating film **600**, **600**A, **600**B: Gate electrode wiring portion **602**, **602**A, **602**B: Gate electrode portion **604**: Gate electrode wiring lead-out portion **605**: Gate electrode lead-out portion **700***a*: Source contact via **700***b*: Drain contact via **700***c*: Body contact via **702**: Silicide film **800***a*: Source electrode **800***b*: Drain electrode **800***c*: Body contact electrode **900**: Silicon oxide film **902**: Silicon nitride film **904**: Resist pattern **906**: Opening **907**: Silicon oxide film **908**: Silicon thin film portion **1000**: Silicon oxide film **1100**: Silicon nitride

film **1200**: Resist film **1300**: Resist pattern **1600**: Opening **1400**: Silicon oxide film **1500**: Resist film **1700**: Transistor formation region **350***a*′, **350***b*′: Resist pattern including protruding portion **2000**: LDD layer **305***a*: Source N+ region **305***b*: Drain N+ region

Claims

1. A semiconductor apparatus, comprising: a substrate having a buried insulating film and a semiconductor layer provided on the buried insulating film and formed with a semiconductor element; a gate electrode provided on the semiconductor layer and having a wiring extending from a central portion of the semiconductor layer to each end portion of the semiconductor layer in a case where the substrate is viewed from above; a source contact via and a drain contact via which are provided on the semiconductor layer; and a protruding portion provided near an area where the end portion of the semiconductor layer intersects the wiring of the gate electrode, the protruding portion including a material identical to that of the semiconductor layer and protruding outward from a side surface of the semiconductor layer, wherein at least part of the end portion of the semiconductor layer has thick film regions each having a larger film thickness than a portion of the semiconductor layer immediately below the gate electrode, and the source contact via and the drain contact via are provided in areas other than the thick film regions.
2. The semiconductor apparatus according to claim 1, wherein the wiring has lead-out portions extending outward from respective end portions of the semiconductor layer, and two or more protruding portions are provided interposing the lead-out portion of the gate electrode between the two or more protruding portions.
3. The semiconductor apparatus according to claim 1, wherein the wiring has lead-out portions extending outward from respective end portions of the semiconductor layer, the lead-out portions of the gate electrode are disposed in plural in such a manner that the lead-out portions are parallel to one another, and the protruding portion is provided at an end portion opposite to an end portion at which the lead-out portions of the gate electrode face each other.
4. The semiconductor apparatus according to claim 1, further comprising: an element isolation portion isolating the semiconductor element.
5. The semiconductor apparatus according to claim 1, wherein the end portion of the semiconductor layer has a film thickness twice to ten times as large as a film thickness of the central portion of the semiconductor layer.
6. The semiconductor apparatus according to claim 5, wherein the end portion of the semiconductor layer has a film thickness of 140 nm to 200 nm, and the central portion of the semiconductor layer has a film thickness of 20 nm to 70 nm.
7. The semiconductor apparatus according to claim 1, wherein the protruding portion protrudes 0.4 μm or more in a direction parallel to an extension direction of the wiring of the gate electrode and protrudes 0.7 μm or more in a direction perpendicular to the extension direction, or the protruding portion protrudes 1 μm or more in the direction parallel to the extension direction and protrudes 1 μm or more in the direction perpendicular to the extension direction.
8. The semiconductor apparatus according to claim 2, wherein a distance between the protruding portion and the lead-out portion of the gate electrode is 14 nm or more and 200 nm or less.
9. The semiconductor apparatus according to claim 1, wherein the protruding portion has at least one shape of a quadrangle, a round shape, a triangle, and a polygon, as viewed from above the substrate.
10. The semiconductor apparatus according to claim 1, wherein the gate electrode has any one of an H shape, a T shape, an I shape, and a ladder shape, as viewed from above the substrate.
11. A manufacturing method for a semiconductor apparatus, the manufacturing method comprising: forming a gate electrode on a semiconductor layer provided on a buried insulating film and formed with a semiconductor element, the gate electrode having a wiring extending from a central portion

of the semiconductor layer to end portions of the semiconductor layer in a case where a substrate having the buried insulating film and the semiconductor layer is viewed from above; forming protruding portions each located near an area where the end portion of the semiconductor layer intersects the wiring of the gate electrode, the protruding portion including a material identical to that of the semiconductor layer and protruding outward from a side surface of the semiconductor layer; and setting the end portions of the semiconductor layer larger in film thickness than a portion of the semiconductor layer immediately below the gate electrode.